

PROJECT-2

**Product Optimization &
Revenue Contribution
Analysis for Afficionado
Coffee Roasters**

PROJECT REPORT

Afficionado Product Analytics: Optimizing Menu Performance and Revenue Impact

Summary:

This project delivers a dynamic Power BI dashboard that analyzes product performance across store locations for Afficionado Coffee Roasters. It highlights revenue contribution, sales volume, category dependency, and product efficiency using interactive filters and visual KPIs. The dashboard empowers stakeholders to identify top-selling items, underperformers, and opportunities for menu simplification and pricing strategy.

Key Performance Indicators (KPIs):

- **Total Revenue:** ₹698K — overall revenue generated from all products.
 - **Total Unit Price:** ₹504.34K — cumulative unit price across all transactions.
 - **Total Quantity:** 214K — total number of units sold across all products.
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Revenue by Product Category

Problem

Most of the revenue is coming from only a few categories.

Insight

- Coffee brings in the most money (₹0.27M), followed by Tea (₹0.19M).
- Bakery and Drinking Chocolate add some revenue but not much.
- Categories like Branded, Loose Tea, Flavours, and Packaged Chocolate hardly make any money.

Recommendations

- Keep focusing on Coffee and Tea since they earn the most.
- Think about reducing or improving low-earning items like Branded and Packaged Chocolate.

- Promote Coffee Beans more because they earn good money even with fewer sales.
 - Try adding new products or seasonal items to spread revenue across more categories.
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Quantity Sold by Product Category

Problem

Sales are mostly concentrated in Coffee and Tea, while other items sell very little.

Insight

- Coffee (89K units) and Tea (70K units) are the most popular.
- Bakery and Drinking Chocolate sell a fair amount.
- Flavours, Loose Tea, Branded, and Packaged Chocolate sell very few or none.
- Coffee Beans sell only 2K units but still make decent revenue, showing they are priced higher.

Recommendations

- Make sure Coffee and Tea are always well stocked since they sell the most.
 - Check why items like Packaged Chocolate and Branded products are not selling maybe remove them or improve them.
 - Run promotions or discounts to increase sales of less popular items.
 - Keep Coffee Beans as a premium product but promote them more to boost sales.
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Relationship Between Revenue and Quantity Sold

Problem

It's not always clear which products are both popular and profitable. Some items may sell a lot but earn less, while others sell less but earn more.

Insight

- The scatter plot shows a **positive correlation** — as quantity increases, revenue also tends to increase.
- Most products follow this trend, but a few points may stand out as exceptions.

- Products with high quantity but low revenue might be underpriced.
- Products with low quantity but high revenue might be premium items.

Recommendations

- **Review pricing** for high-selling items that don't bring in much revenue.
 - **Promote premium products** that earn more even with fewer sales.
 - **Use this chart** to find products that are both popular and profitable — these are your best performers.
 - **Look closer at outliers** to decide if they need price changes, better marketing, or removal.
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Revenue Distribution by Product Type

Problem

It's hard to know which specific product types are driving most of the revenue. Without this breakdown, decisions on pricing, promotions, or menu design may be less effective.

Insight

- A few product types like Barista Espresso, Brewed Chai Tea, and Hot Chocolate take up the largest space in the treemap — meaning they generate the most revenue.
- Other types like Gourmet Brewed Coffee, Organic Brewed Coffee, and Drip Coffee also contribute well.
- Smaller blocks like Biscotti, Scone, Clothing, and Housewares show very low revenue, suggesting limited demand or poor placement.
- Some premium items like Espresso Beans and Premium Beans may be priced higher but sell less.

Recommendations

- **Focus on top earners** like Barista Espresso and Brewed Chai Tea — keep them well stocked and promote them.
- **Review low-revenue items** like Clothing and Housewares — consider removing or replacing them.

- **Group similar products** (e.g., all brewed teas) to simplify the menu and improve customer experience.
- **Highlight premium items** like Espresso Beans with better placement or combo offers.
- **Use this treemap** to guide which products deserve more attention in marketing and inventory planning.

SOLVING BUSINESS QUESTIONS USING SQL QUERIES

Problem

- Some products are sold in very large quantities.
- But they don't generate much total revenue.
- This could lead to low profit margins or missed business opportunities.

Goal

Find products that are **popular in terms of quantity sold** but **earn less money overall**. These are high-volume, low-revenue items that may need pricing or strategy adjustments.

Logic Behind the Query

- For each product, calculate how many units were sold.
- Then calculate how much total money that product earned.
- Sort the results to show products with **highest quantity** and **lowest revenue** first.

Insights from Your Data

- **Brewed Black Tea** and **Drip Coffee** are sold in large numbers but earn less revenue per unit.
- **Barista Espresso** sells in similar volume but earns more, showing better pricing or value.
- Some **Tea** and **Chocolate** items have high volume but low revenue, which may indicate low margins or heavy discounts.

Recommendations in Simple Language

- **Check prices:** Popular items might be too cheap. Raising the price a little could increase revenue.
 - **Look at costs:** If making or serving these items is expensive, they may not be worth selling in such high numbers.
 - **Change promotions:** Don't give too many discounts on items that already sell a lot but don't earn much.
 - **Bundle smartly:** Combine low-earning items with high-profit ones (like selling tea with a snack) to balance revenue.
 - **Show the data visually:** Use charts (like quantity vs revenue) so managers can quickly see which products are "high sales but low money."
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SOLVING BUSINESS QUESTIONS USING SQL QUERIES

Problem

- Some products are priced very high.
- But they are sold only a few times.
- This could mean customers don't find them valuable or affordable.

Goal

Find products that have **high prices** but **low sales counts**. These items may need better marketing, pricing adjustments, or replacement.

Logic Behind the Query (in simple words)

- For each product, check its price.
- Count how many times it was sold.
- Sort the results to show **most expensive** items with **least sales** at the top.

Insights from Your Data

- Items like **Premium Beans** (₹45, sold 190 times) and **Clothing** (₹28, sold 216 times) are expensive but not sold very frequently.
- Some products like **Housewares** and **Organic Beans** priced above ₹20 are sold only 5–11 times.
- This shows that high price doesn't always mean high demand.

Recommendations in Simple Language

- **Check if the price is too high:** Maybe customers aren't willing to pay that much.
 - **Improve product visibility:** Are these items easy to find or attractive to buyers?
 - **Offer discounts or bundles:** Try selling them with popular items to increase interest.
 - **Review product quality or usefulness:** Are they worth the price from a customer's view?
 - **Use charts:** A bar chart showing price vs sales count can help managers spot weak performers quickly.
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Problem

- Some product categories have too many different items.
- This can confuse customers and slow down service.
- It may also make inventory and menu management harder.

Goal

Find which product categories have too many variants (different items). These are overcrowded sections that may need simplification.

Logic Behind the Query (in simple words)

- For each product category, count how many unique products it includes.
- This shows how “crowded” each category is.
- Sort the results to show categories with the most items at the top.

Insights from Your Data

- **Coffee** has 21 different items — the most crowded category.
- **Tea** and **Bakery** follow with 16 and 11 items.
- Smaller categories like **Packaged Chocolate, Flavours**, and **Branded** have only 3–4 items, which are easier to manage.

Recommendations in Simple Language

- **Simplify crowded categories:** Reduce similar items in Coffee and Tea to speed up service.
 - **Group similar products:** Combine variants under one menu option if differences are minor.
 - **Highlight bestsellers:** Keep only top-performing items to reduce decision time.
 - **Use clear menu design:** Make it easy for staff and customers to find what they want.
 - **Monitor performance:** Track which items are rarely ordered and consider removing them.
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Problem

- The business offers many products, but not all are equally popular.
- Some items are purchased much more frequently than others.
- Knowing which products customers prefer helps improve sales and service.

Goal

Find the **top 10 products** that are purchased most often. These are customer favorites and can guide decisions on promotions, inventory, and menu design.

Logic Behind the Query (in simple words)

- Count how many times each product was purchased.
- Group the results by product ID, category, and type.
- Show the top 10 products with the **highest purchase counts**.

Insights from Your Data

- **Pastry, Brewed Black Tea, and Hot Chocolate** are among the most purchased items.
- **Gourmet brewed coffee** and **Barista Espresso** also rank high in customer preference.
- Multiple variants of **Chai tea** and **Hot chocolate** appear in the top 10, showing strong demand across similar types.

Recommendations in Simple Language

- **Keep these items well-stocked:** They're customer favorites and drive repeat purchases.
 - **Promote them more:** Highlight these popular items in menus or offers.
 - **Use them in combos:** Pair them with less popular items to boost overall sales.
 - **Monitor quality and speed:** Ensure these high-demand items are served quickly and consistently.
 - **Design menus around them:** Make them easy to find and order to improve customer experience.
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Problem

- The business sells many products, but not all contribute equally to revenue.
- Some items generate much more money than others.
- Knowing which products earn the most helps focus on what drives profit.

Goal

Identify the top 10 products that bring in the highest total revenue. These are the most valuable items for the business and should be prioritized in strategy and planning.

Logic Behind the Query (in simple words)

- For each product, calculate how much money it earned by multiplying quantity sold with its price.
- Group the results by product ID, category, and type.
- Show the top 10 products with the highest total revenue.

Insights from Your Data

- **Hot Chocolate** and **Barista Espresso** are top revenue generators, appearing multiple times with high totals.
- **Brewed Chai Tea** and **Premium brewed coffee** also perform strongly.
- Most high-revenue products come from **Coffee** and **Drinking Chocolate** categories, showing strong customer demand and pricing efficiency.

Recommendations in Simple Language

- **Keep these items well-stocked:** They bring in the most money.
 - **Promote them more:** Use them in ads, combos, or loyalty programs.
 - **Ensure quality and speed:** These products should be served consistently to maintain customer satisfaction.
 - **Use them in planning:** Focus on these items when designing menus or forecasting revenue.
 - **Monitor trends:** Track if their performance changes over time to adjust strategy quickly.
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Problem

The business offers many products, but only a few are sold in very large quantities. Identifying these helps focus on what customers buy most often.

Goal

Find the top 10 products with the highest total quantity sold. These are the most popular items and key drivers of daily operations.

Query Logic (in simple words)

- Add up how many units were sold for each product.
- Group by product ID, category, and type.
- Show the top 10 products with the highest total quantity sold.

Insights from Your Data

- Brewed Black Tea, Hot Chocolate, and Barista Espresso are among the top-selling items.
- Tea and Coffee categories dominate the list, showing strong customer preference.
- These products likely drive high footfall and repeat orders.

Recommendations

- Keep these items well-stocked to avoid missed sales.
- Ensure fast service for these popular products to maintain customer satisfaction.
- Use them in promotions or combos to boost overall sales

Monitor trends to see if their popularity changes over time.

Problem

Some products are not selling well. They have very low total quantity sold, which could mean they're not popular, not visible, or not useful to customers.

Goal

Identify the **bottom 10 products** with the **lowest total quantity sold**. These are the least-selling items and may need review or removal.

Query Logic (in simple words)

- Add up how many units were sold for each product.
- Group by product ID, category, and type.
- Show the 10 products with the **lowest total quantity sold**.

Insights from Your Data

- Products like **Drinking Chocolate**, **Loose Tea**, and **Coffee Beans** appear frequently in the least-selling list.
- Most of these items sold fewer than 160 units, which is very low compared to top-selling items.
- This may indicate low demand, poor visibility, or pricing issues.

Recommendations in Simple Language

- **Review these products:** Are they needed on the menu or shelf?
 - **Check visibility:** Are they easy for customers to find or order?
 - **Consider removing or replacing:** If they don't sell, they may be wasting space.
 - **Try promotions:** Offer discounts or bundles to test if demand improves.
 - **Monitor performance:** Track if sales improve after changes, or if they stay low.
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Problem

Some products are purchased very frequently (popular), but they may not earn much money (profitable). Others might be less popular but generate high revenue. This mismatch can affect business decisions and profit margins.

Goal

Compare **how often products are purchased** (popularity) with **how much money they earn** (profitability). This helps identify:

- High-demand but low-profit items
- Low-demand but high-profit items
- Ideal products that are both popular and profitable

Query Logic (in simple words)

- Count how many times each product was purchased → this shows popularity
- Calculate total revenue earned from each product → this shows profitability
- Group by product ID, category, and type
- Sort to show products with **most purchases** and **highest revenue** first

Insights from Your Data

- **Pastry, Brewed Black Tea, and Hot Chocolate** are highly popular and also profitable — ideal performers
- **Barista Espresso** and **Gourmet brewed coffee** also rank well in both popularity and profitability
- Some products like **Loose Tea** and **Packaged Chocolate** are less popular and earn less, indicating low impact

Recommendations in Simple Language

- **Focus on products that are both popular and profitable** — they drive business success
- **Review pricing** for popular but low-profit items — they might be underpriced
- **Promote profitable but less popular items** — use combos or better placement
- **Use charts** to compare popularity vs profitability — a scatter plot can show which products are in the “sweet spot”
- **Balance the menu** — keep a mix of crowd-pleasers and high-margin items to optimize both sales and profit

